

Practical 1: Designing and Implementing a Multi-Zone (DMZ)

INTRODUCTION

This practical focuses on the design and implementation of a secure multi-zone network architecture using Cisco Packet Tracer. A three-tier model consisting of the Wide Area Network (WAN), Demilitarized Zone (DMZ), and Local Area Network (LAN) was developed to simulate a real-world enterprise environment. Access Control Lists (ACLs) were configured on a router to enforce security policies, regulate traffic flow between the zones, and ensure controlled communication across the network.

OBJECTIVE

The objective of this practical is to design and implement a secure network architecture with three distinct zones: WAN, DMZ, and LAN. It aims to configure a router using Access Control Lists (ACLs) to control and restrict traffic between these zones while allowing only necessary communication. Additionally, the practical seeks to protect sensitive resources, such as the database server, from direct external access while maintaining accessibility to public services like the web server.

PHYSICAL TOPOLOGY SETUP

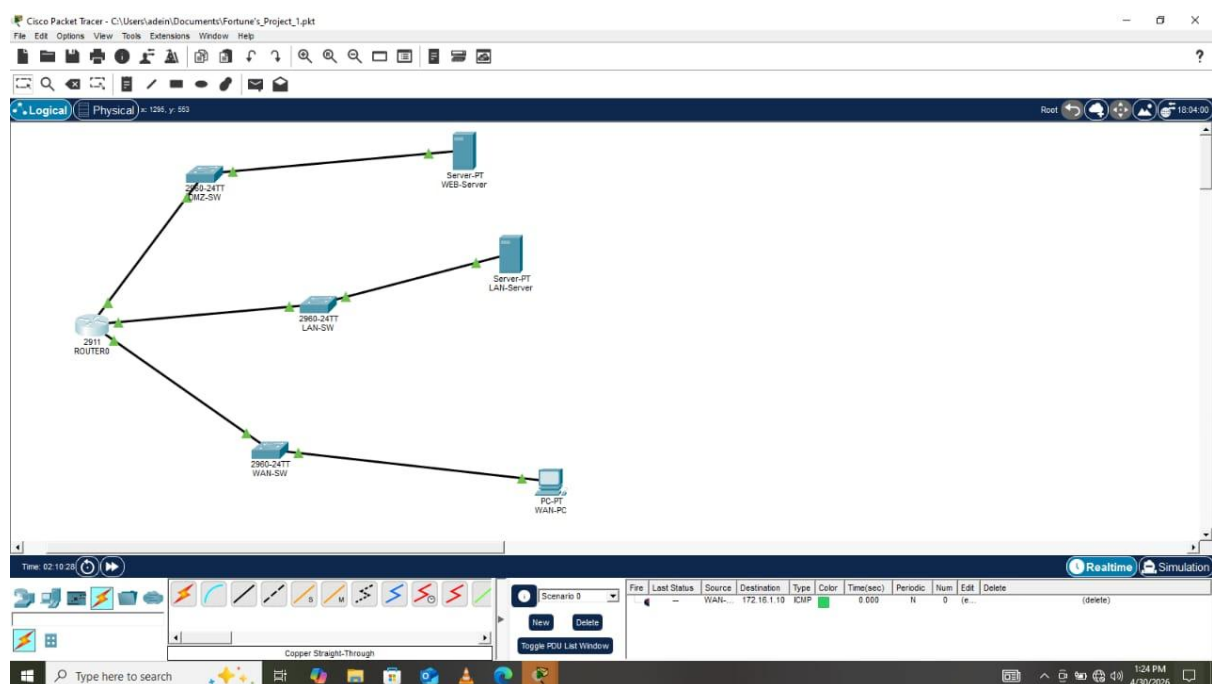
To build this network in Cisco Packet Tracer, the following steps were taken:

Router:

The Network Devices icon was selected and a 2911 Router was placed at the center of the workspace.

Switches:

Two 2960-24TT switches were added. One was placed above the router (WAN switch) and one below (LAN switch). A third switch was used for the DMZ.



End Devices:

A PC (WAN PC) was placed at the top

A Server (DMZ Web Server) was placed in the middle

A PC (LAN PC) and another Server (LAN Database) were placed at the bottom

CABLING

Connections were made using Copper Straight-Through cables:

WAN PC → WAN Switch

WAN Switch → Router (G0/0)

DMZ Server → Router (G0/1)

LAN Server → Router (G0/2)

The WAN, DMZ, and LAN zones were all connected to the router positioned at the center.

IP ADDRESSING SCHEME/DESCRIPTION

Network Zone	IP Range	Device	IP Address
WAN	192.168.1.0/24	Attacker (Kali Linux)	192.168.1.10
WAN	192.168.1.0/24	Firewall (Outside Interface)	192.168.1.1
DMZ	172.16.1.0/24	Firewall (DMZ Interface)	172.16.1.1
DMZ	172.16.1.0/24	Web Server	172.16.1.10
LAN	10.0.1.0/24	Firewall (Inside Interface)	10.0.1.1
LAN	10.0.1.0/24	Database Server	10.0.1.10

CONFIGURATION

THIS IS THE POINT WHERE THE ROUTER IS CONFIGURED BY ALLOCATING IP ADDRESSES TO EACH INTERFACES

ROUTER INTERFACE CONFIGURATION

interface g0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

interface g0/1

ip address 172.16.1.1 255.255.255.0

no shutdown

interface g0/2

ip address 10.0.1.1 255.255.255.0

no shutdown

Rule Configuration

Access Control Lists (ACLs) were configured using the router CLI

Router> enable

Router# configure terminal

RULE A

ALLOW HTTP/HTTPS from WAN to DMZ Web server

```
access-list 100 permit tcp 192.168.1.0 0.0.0.255 host 172.16.1.10  
eq 80
```

```
access-list 100 permit tcp 192.168.1.0 0.0.0.255 host 172.16.1.10  
eq 443
```

Rule B: Allow SQL traffic from DMZ to LAN Database

```
access-list 100 permit tcp host 172.16.1.10 host 10.0.1.20 eq  
1433
```

Rule C: Deny all traffic from WAN to LAN

```
access-list 100 deny ip 192.168.1.0 0.0.0.255 10.0.1.0 0.0.0.255
```

Rule D: Deny all traffic from DMZ to WAN

```
access-list 100 deny ip 172.16.1.0 0.0.0.255 192.168.1.0  
0.0.0.255
```

Default Rule

```
access-list 100 permit ip any any
```

Apply ACL to the Router Interface Configuration

interface g0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

interface g0/1

ip address 172.16.1.1 255.255.255.0

no shutdown

interface g0/2

ip address 10.0.1.1 255.255.255.0

no shutdown

SECURITY POLICY IMPLEMENTATION

Allow HTTP/HTTPS from WAN to Web Server (DMZ):
HTTP(S) traffic from WAN to Web Server was allowed
through port 80 and 443

Allow SQL traffic from Web Server to Database:
Traffic was allowed through port 1433

Deny all traffic from WAN to LAN: All traffic were
denied from WAN to LAN

Deny traffic from DMZ to WAN: All traffic were denied from DMZ to WAN

Allow all other safe traffic: All traffics not included in the ACL were allow

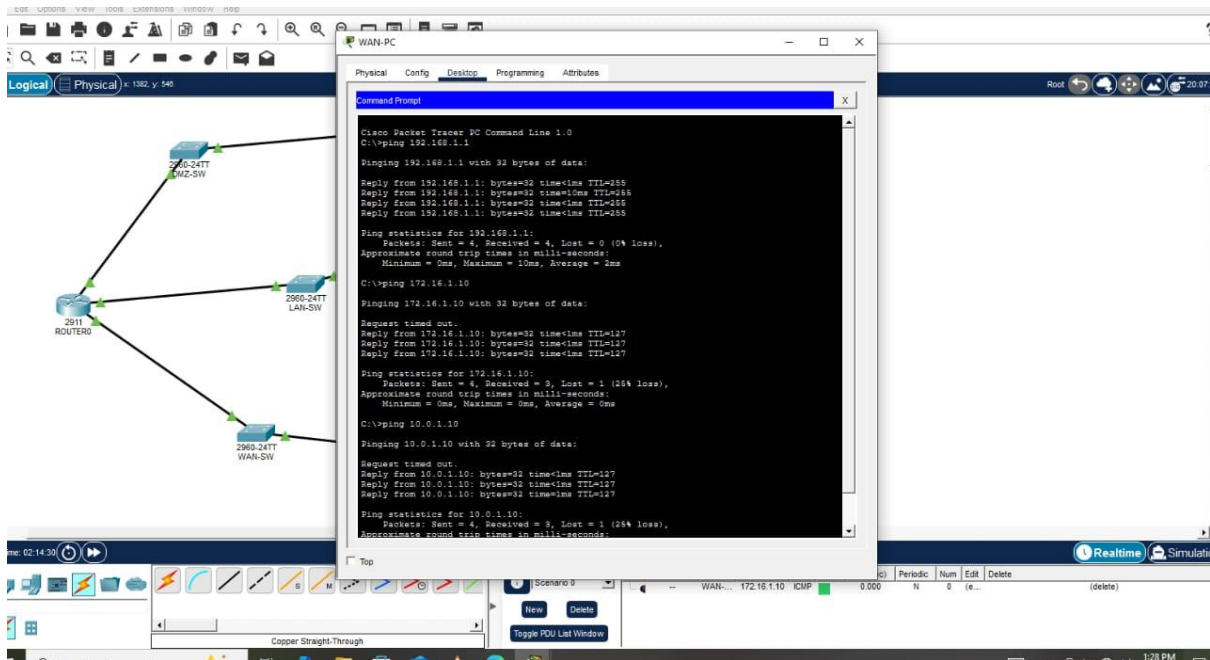
TESTING

Connectivity was tested by pinging the DMZ server from the WAN PC:

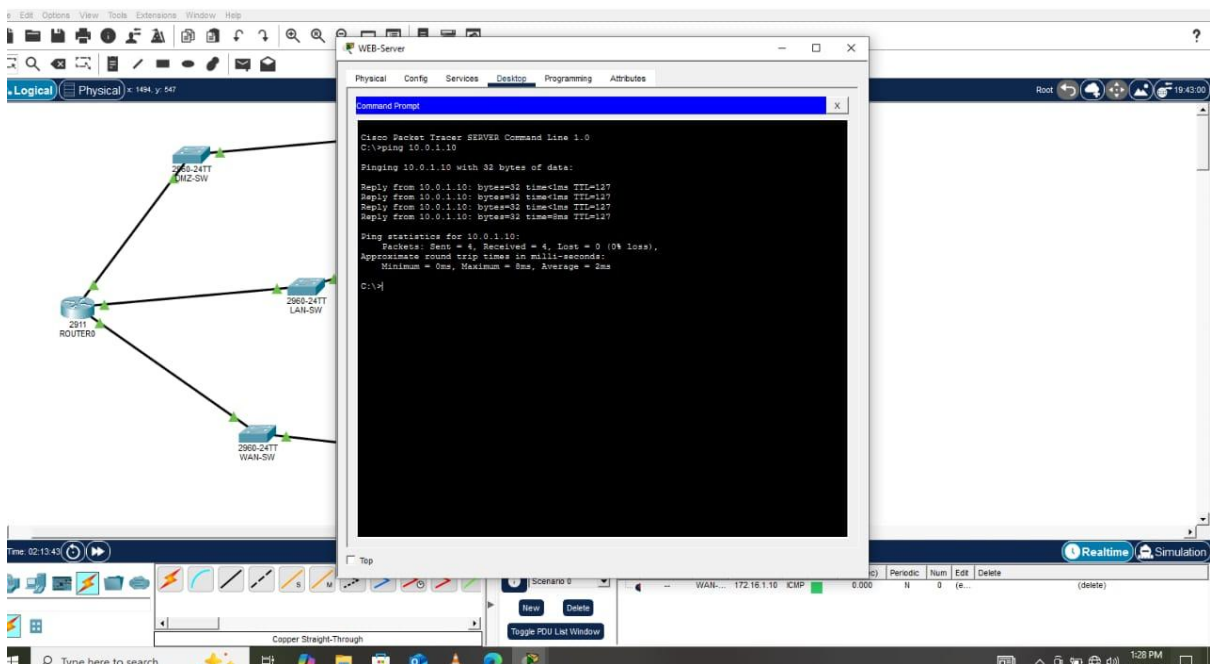
ping 172.16.1.10

The request was successful, confirming allowed communication.

Test 1: DMZ to WAN All traffic from DMZ to WAN were denied as shown below



Test 2: WAN to Database Server Direct access from
 WAN to LAN database server was blocked,
 confirming proper security implementation as
 shown below;



VERIFICATION

Due to limitations in Cisco Packet Tracer, port scanning tools like Nmap were unavailable.

Verification was carried out using:

```
show access-lists 100
```

This confirmed that HTTP (80) and HTTPS (443) traffic were permitted while other traffic remained restricted.

RESULTS

The implemented ACL successfully enforced network segmentation and security policies. Public users were able to access the web server, while direct access to the database server was restricted. The system effectively prevented unauthorized access and ensured controlled communication between network zones.

CONCLUSION

This practical demonstrated how network segmentation and ACLs can be used to secure enterprise environments. By implementing a multi-zone architecture, sensitive systems were protected while still allowing necessary communication. This approach is widely used in real world network security.